10.18 Dimensional Regularization and the $\overline{\text{MS}}$ Scheme

The Core Mechanism of Dimensional Regularization

In dimensional regularization (dim reg), we analytically continue the spacetime dimension from $d = 4$ to $d = 4 - \epsilon$ (some books use $d = 4 - 2\epsilon$; it is a matter of convention). The fundamental quantity governing divergences is the Euler Gamma function, as seen in the master formula for Euclidean integrals:

$$\int \frac{d^d k}{(2\pi)^d} \frac{1}{(k^2 + \Delta)^n} = \frac{1}{(4\pi)^{d/2}} \frac{\Gamma(n - \frac{d}{2})}{\Gamma(n)} \left(\frac{1}{\Delta}\right)^{n-d/2} \tag{10.18.1}$$

The Gamma function $\Gamma(z)$ has simple poles at non-positive integers $z = 0, -1, -2, \ldots$. Therefore, divergences (as poles in ϵ) appear when the argument $n - d/2$ approaches one of these integers. For one-loop integrals of this form the Gamma-function poles are the *only* source of ultraviolet divergences: if $\Gamma(n - d/2)$ is finite, the integral is finite in this scheme.

Odd versus Even Dimensions

The behavior of loop integrals differs between odd and even dimensions. For a one-loop diagram ($L = 1$) with I internal scalar propagators and no derivative couplings, the superficial degree of divergence is

$$\omega = dL - 2I = d - 2I.$$

Look at its parity:

- If d is even, $\omega = (\text{even}) - 2I$ is even, so $\omega = 0$ (a logarithmic divergence) is possible.
- If d is odd, $\omega = (\text{odd}) - 2I$ is *always* odd.

So, by power counting, a one-loop diagram in an odd dimension cannot be logarithmically divergent. This explains the "special" nature of odd dimensions in dim reg:

- **Even dimensions ($d = 2, 4, 6, \ldots$):** $d/2$ is an integer. One-loop diagrams can have $\omega = 0, 2, 4, \ldots$. The Gamma-function argument $n - d/2$ can hit a pole, producing $1/\epsilon$ divergences for *both* power-law and logarithmic cases.

- **Odd dimensions ($d = 3, 5, 7, \ldots$):** $d/2$ is a half-integer. One-loop diagrams can only have $\omega = 1, 3, 5, \ldots$. The Gamma-function argument $n - d/2$ is *always* a half-integer, which is *never* a pole. Thus **all one-loop diagrams in odd dimensions are finite in dim reg**. But we know a divergence is a divergence; it does not disappear. Dim reg does something special, explained below.

Clarification: this only applies at one loop. In odd dimensions, logarithmic divergences appear at even loop orders, where $\omega = Ld - 2I$ can vanish. In ϕ^6 theory in $d = 3$, the one-loop diagrams are finite, but at two loops $\omega = 2d - 2I = 0$ for $I = 3$, and dim reg shows the corresponding $1/\epsilon$ pole. This is why renormalization is still needed in odd dimensions.

Power-Law versus Logarithmic Divergences

Dim reg treats the two kinds of divergence very differently.

- **Scaleless integrals**, which have no mass parameter at all, like $\int d^dk\,(k^2)^a$, are analytically continued to zero. This is a built-in feature of the scheme. Introducing the renormalization scale μ ($d^dk \to \mu^{4-d}d^dk$, to keep couplings dimensionless) does not change this: it cannot generate a mass scale inside the integral.
- **Integrals with a mass scale** $\Delta = m^2$ keep only the part proportional to powers of m.

In $d = 4 - \epsilon$:

- **Logarithmic ($\omega = 0$):** $I_{\log} = \int \frac{d^dk}{(k^2+\Delta)^2} \propto \Gamma(2 - d/2) = \Gamma(\epsilon/2) = \frac{2}{\epsilon} - \gamma_E + \mathcal{O}(\epsilon)$. This is a "pure" $1/\epsilon$ pole.
- **Power law ($\omega = 2$):** $I_{\text{quad}} = \int \frac{d^dk}{k^2+\Delta} \propto \Gamma(1 - d/2) = \Gamma(-1 + \epsilon/2) \sim \mathbf{m^2} \times \frac{1}{\epsilon}$. The m^2 comes from the factor $(1/\Delta)^{n-d/2}$ in Eq. (10.18.1), which becomes $\Delta^{d/2-1} \approx \Delta = m^2$.

What does the m^2/ϵ pole correspond to? Compare a Euclidean momentum cutoff Λ:

$$\int^{\Lambda} \frac{d^4k}{(2\pi)^4} \frac{1}{k^2 + m^2} = \frac{1}{16\pi^2}\left[\Lambda^2 - m^2 \ln \frac{\Lambda^2}{m^2} + \mathcal{O}\left(\frac{m^4}{\Lambda^2}\right)\right] \quad \text{(cutoff)},$$

$$\int \frac{d^dk}{(2\pi)^d} \frac{1}{k^2 + m^2} = -\frac{m^2}{16\pi^2}\left[\frac{2}{\epsilon} + \ldots\right] \quad \text{(dim reg)}.$$

The quadratically divergent integral contains a subleading *logarithmic* divergence, $-m^2 \ln \Lambda^2$, and that is what the pole is: $\ln \Lambda^2 \leftrightarrow 2/\epsilon$. The pure power Λ^2 has no counterpart in dim reg; it is simply dropped. So it is tempting to say that in even dimensions dim reg "tags" the pole with the

mass dimension of the power-law divergence, but more precisely, **every $1/\epsilon$ pole is a logarithmic divergence**, multiplied by whatever powers of the masses the dimensions require. In odd dimensions there is no logarithm at one loop, and a power-divergent integral is evaluated to a finite multiple of m^ω.

Dim reg is subtle: it discards every pure power-law divergence. For a scaleful integral (with masses in Δ) what is left of a divergence of degree ω is m^ω times a $1/\epsilon$ pole (a logarithm) in even dimensions, or times a finite number in odd dimensions. For a scaleless integral nothing is left. This automatic subtraction of pure power divergences is both a feature (computational simplicity) and a limitation (it hides the sensitivity of the results to high scales).

The $\overline{\text{MS}}$ Scheme: What Are We Really Doing?

In the $\overline{\text{MS}}$ (modified minimal subtraction) scheme, we:

1. Subtract **only** the $1/\epsilon$ poles (together with the constants $\gamma_E - \ln(4\pi)$ that accompany them).
2. Do *not* "ignore" power-law divergences. Rather, we **implicitly accept dim reg's evaluation of them**, finite or zero.

This works because in a renormalizable QFT all power-law divergences can be absorbed by the same counterterms that absorb the logarithmic ones. The $\overline{\text{MS}}$ scheme appears "minimal" only because the regulator has already done the heavy lifting of removing $\Lambda^2, \Lambda^3, \ldots$.

Contrast with Pauli–Villars regularization:

$$\Pi_{\text{PV}}(p^2) \sim c_1\Lambda^2 + c_2 m^2 \ln\left(\frac{\Lambda^2}{\mu^2}\right) + \text{finite}.$$

Here we explicitly see both quadratic (Λ^2) and logarithmic ($\ln\Lambda^2$) divergences. A counterterm must be "brute-forced" to cancel the Λ^2 term explicitly. In $\overline{\text{MS}}$, the Λ^2 term never appears.

Only the $1/\epsilon$ Poles Drive the Running

The renormalization-group flow, described by the β-function $\beta(g_R) = \mu\, dg_R/d\mu$, tells us how a coupling g_R must change with the scale μ to keep physical observables independent of our choice of μ. The bare coupling g_0 is independent of μ. In $d = 4 - \epsilon$ dimensions,

$$g_0 = \mu^\epsilon \left(g_R + \frac{a_1(g_R)}{\epsilon} + \ldots \right).$$

Setting $\mu\, dg_0/d\mu = 0$ and keeping the ϵ^0 terms gives

$$\beta(g_R) = \left(g_R \frac{d}{dg_R} - 1 \right) a_1(g_R),$$

so β is determined *entirely* by a_1, the residue of the $1/\epsilon$ pole. For $\lambda\phi^4/4!$ theory, $a_1 = 3\lambda^2/16\pi^2$ and $\beta = 3\lambda^2/16\pi^2$.

The same holds for the mass. The m^2/ϵ pole of the quadratically divergent tadpole is a genuine $1/\epsilon$ residue, and it drives the running of m^2: in $\lambda\phi^4/4!$ theory, $m_0^2 = m^2\big(1 + \lambda/16\pi^2\epsilon + \dots\big)$, so that $\mu\, dm^2/d\mu = \lambda m^2/16\pi^2$ at one loop. What does *not* run is the pure power Λ^2, which dim reg has already discarded. So the precise statement is: the running is fixed by the logarithmic divergences, including the logarithms that sit inside power-divergent integrals with mass-dimension prefactors. The finite parts left after subtraction are not μ-independent: they carry $\ln(\mu^2/m^2)$ and $\ln(\mu^2/p^2)$, and their μ-dependence is exactly what the running of the couplings compensates.

Mass Independence, Thresholds and Decoupling

Because $\overline{\text{MS}}$ subtracts only the $1/\epsilon$ poles, and their coefficients are polynomials in the masses with mass-independent coefficients, the renormalization factors Z (and the counterterm coefficients δ_m, $\delta_\lambda, \dots$) do not depend on the masses. The β-functions of the dimensionless couplings are therefore mass independent: the scheme is "blind" to the masses of the particles in the loops.

The physical consequence is that $\overline{\text{MS}}$ **does not see particle thresholds**. When the scale drops below the mass M of a heavy particle, that particle should stop affecting low-energy physics (the Appelquist–Carazzone decoupling theorem, 1975). In a physical, momentum-subtraction scheme this happens automatically: the heavy particle's contribution to the running of the light couplings switches off for $\mu \ll M$. In $\overline{\text{MS}}$ the heavy particle keeps contributing to the β-functions at all μ, and its effect at low energy reappears as large logarithms $\ln(M/\mu)$ in the finite parts. One restores decoupling by hand: integrate the heavy particle out at $\mu \sim M$ and match the full theory to an effective field theory without it.

Example: in the Standard Model the top quark ($m_t \approx 173$ GeV) should not affect the running of the light couplings at $\mu \ll m_t$. In $\overline{\text{MS}}$ this has to be implemented through decoupling relations: for instance, the strong coupling below m_t is run with five quark flavors after matching at $\mu \sim m_t$.

Physical Implications and the Limits of Mathematical Elegance

While dim reg + $\overline{\text{MS}}$ is computationally optimal, it obscures important physical puzzles by "hiding" power-law divergences.

The Hierarchy Problem

In cutoff regularization, the top-quark loop corrects the Higgs mass by

$$\delta m_H^2 \sim y_t^2 \Lambda^2 + y_t^2 m_t^2 \ln\left(\frac{\Lambda^2}{\mu^2}\right).$$

The quadratic term Λ^2 represents a dramatic fine-tuning problem if $\Lambda \sim M_{\rm Pl}$. In dim reg,

$$\delta m_H^2 \sim y_t^2 m_t^2 \left(\frac{1}{\epsilon} + \text{finite}\right).$$

The Λ^2 divergence is gone. The fine-tuning problem becomes mathematically invisible, but physically it remains: if the Higgs couples to any heavy particle of mass M, dim reg gives a correction proportional to M^2, and the sensitivity to the heavy scale is back.

The Cosmological Constant Problem

With a cutoff, the zero-point energy density of a free field is

$$\rho_{\rm vac} = \frac{1}{2}\int^{\Lambda} \frac{d^3k}{(2\pi)^3}\sqrt{\mathbf{k}^2 + m^2} \sim \frac{\Lambda^4}{16\pi^2}.$$

In dim reg, for a massless field this is a scaleless integral, which the rules define to be **zero**; for a massive field only a term proportional to $m^4 \ln(m^2/\mu^2)$ survives. This regularization *deletes* the Λ^4 from the calculation, but it does not remove the problem: the m^4 contributions of the known particles alone exceed the observed vacuum energy by many orders of magnitude.

The Lesson

(As I emphasized in class.) We must distinguish between mathematical convenience and physical insight. Different (reasonable) regularization schemes make different problems or properties of the underlying theory apparent. Dim reg gives the correct answers for renormalized observable quantities (like scattering amplitudes) but may obscure the deeper physical questions about naturalness, UV sensitivity, and decoupling that are made manifest in cutoff-based schemes. The choice of regularization scheme is not just computational: it embodies a philosophy about what constitutes "real" physics. For a bottom-up effective field theorist, a renormalizable theory is self-consistent to arbitrarily high scales, and won't see any problem with the underlying theory. For a top-down UV theorist who wants to understand the ultimate Lagrangian of nature, consistency is only one aspect; calculability (and hence the predictivity of the bare parameters) is crucial, as long as one does not want to admit that we live in a special and

randomly tuned universe. Misinterpretations of the $\overline{\text{MS}}$ scheme have caused a tremendous amount of confusion in the field, even today. I hope you learned something from this note.

Summary

- **Mathematical advantage:** dim reg + $\overline{\text{MS}}$ is the most efficient scheme, preserving symmetries and keeping only the universal logarithmic running.
- **Odd dimensions** are a clean laboratory: one-loop finiteness by power counting, and $1/\epsilon$ poles only at even loop orders ($L = 2, 4, \ldots$) in scalar theories.
- **Every $1/\epsilon$ pole is a logarithm.** Pure power divergences are discarded, and the m^2/ϵ of a quadratically divergent integral is the logarithm inside it; it drives the running of m^2.
- **Physical limitation:** the scheme makes fine-tuning problems (like the hierarchy problem) mathematically invisible, sets scaleless power divergences (like the massless vacuum energy) to zero, and, being mass independent, does not decouple heavy particles by itself.